CPS188: Lab 05 - Pointers and Advanced Functions
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Problem #1:
Algorithm:
1. Ask the user to enter two integer numbers
2. Calculate and output the average, sum of square, absolute value of square difference and square root of sum of square of the two numbers

Code:
```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

void calculate(int num1, int num2, double *avg, int *sumSq, int
*absSqDiff, double *sqrtSumSq) {
    *avg = (num1 + num2) / 2.0; //calculating average
    *sumSq = pow(num1,2) + pow(num2,2); //calculating sum of square
    *absSqDiff = abs((num1 - num2) * (num1 - num2)); //calculating
absolute value of square difference
    *sqrtSumSq = sqrt(*sumSq); //calculating square root of sum of
square
}

int main(void) {
    //initializing variables:
    int num1, num2, sumSq, absSqDiff;
    double avg, sqrtSumSq;

    printf("User, enter two integers: ");
    scanf("%d %d", &num1, &num2);

    //call function
    calculate(num1, num2, &avg, &sumSq, &absSqDiff, &sqrtSumSq);

    printf("Average (rounded to two decimal places): %.2lf\n", avg);
//output average
    printf("Sum of squares: %d\n", sumSq); //output sum of sqaure of
two numbers
```

```
    printf("Absolute value of the square of the difference: %d\n",
absSqDiff); //output absolute value of sqaure difference
    printf("Square root of the sum of the squares (rounded to two
decimal places): %.2lf\n", sqrtSumSq); //output sqaure root of sum of
sqaures

    return (0);
}
```

Output:

```
Set 1:
User, enter two integers: 56 78
Average (rounded to two decimal places): 67.00
Sum of squares: 9220
Absolute value of the square of the difference: 484
Square root of the sum of the squares (rounded to two decimal places):
96.02

Set 2:
User, enter two integers: 45 -20
Average (rounded to two decimal places): 12.50
Sum of squares: 2425
Absolute value of the square of the difference: 4225
Square root of the sum of the squares (rounded to two decimal places):
49.24

Set 3:
User, enter two integers: -80 -40
Average (rounded to two decimal places): -60.00
Sum of squares: 8000
Absolute value of the square of the difference: 1600
Square root of the sum of the squares (rounded to two decimal places):
89.44

Set 4:
User, enter two integers: 8 0
Average (rounded to two decimal places): 4.00
Sum of squares: 64
Absolute value of the square of the difference: 64
```

```
Square root of the sum of the squares (rounded to two decimal places): 8.00
```

Screenshot:

Set 1:

```
User, enter two integers: 56 78
Average (rounded to two decimal places): 67.00
Sum of squares: 9220
Absolute value of the square of the difference: 484
Square root of the sum of the squares (rounded to two decimal places): 96.02


...Program finished with exit code 0
Press ENTER to exit console.
```

Set 2:

```
User, enter two integers: 45 -20
Average (rounded to two decimal places): 12.50
Sum of squares: 2425
Absolute value of the square of the difference: 4225
Square root of the sum of the squares (rounded to two decimal places): 49.24


...Program finished with exit code 0
Press ENTER to exit console.
```

Set 3:

```
User, enter two integers: -80 -40
Average (rounded to two decimal places): -60.00
Sum of squares: 8000
Absolute value of the square of the difference: 1600
Square root of the sum of the squares (rounded to two decimal places): 89.44


...Program finished with exit code 0
Press ENTER to exit console.
```

Set 4:

```
User, enter two integers: 8 0
Average (rounded to two decimal places): 4.00
Sum of squares: 64
Absolute value of the square of the difference: 64
Square root of the sum of the squares (rounded to two decimal places): 8.00


...Program finished with exit code 0
Press ENTER to exit console.
```

Problem #2:

Algorithm:

1. Ask the user to enter one out of the four options.
2. If user chooses an option from 1 to 3 then ask them the speed they would like to travel and output the minimum and maximum time in hours it will take them to travel to the location they chose to travel.
3. Continuously repeat the above steps until the user chooses option 4, to exit the program.

Code:

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

//Travelling to Moon
void moon(double speed, double *minimum, double *maximum){
    double perigee=363104;
    double apogee=405696;
    *maximum=apogee/speed;
    *minimum=perigee/speed;
}

//Travelling to Mars
void mars(double speed, double *minimum, double *maximum){
    double perigee=54600000;
    double apogee=401000000;
    *maximum=apogee/speed;
    *minimum=perigee/speed;
}

//Travelling to Venus
void venus(double speed, double *minimum, double *maximum){
    double perigee=8000000;
    double apogee=261000000;
    *maximum=apogee/speed;
    *minimum=perigee/speed;
}


int main(void) {
    int option;
    double speed, minimum, maximum;
```

```
    do{
        printf("Pick one of the following options by entering the
number corresponding to the option: \n1. Travel to the Moon \n2.
Travel to Mars \n3. Travel to Venus \n4. Exit Program\nOption: ");
//options
        scanf("%d", &option);
        if(option>=1 && option <=3){

            do{
                printf("Enter your speed in kilometers per hour: ");
                scanf("%lf", &speed);
            } while (speed<=0);
        }
        if(option==1){
            moon(speed,&minimum, &maximum);
            printf("It will take you %0.2lf to %0.2lf hours to reach
the Moon from Earth at the speed of %0.2lf km/h.\n\n", speed, minimum,
maximum);
        }
        if(option==2){
            mars(speed,&minimum, &maximum);
            printf("It will take you %0.2lf to %0.2lf hours to reach
the Mars from Earth at the speed of %0.2lf km/h.\n\n", speed, minimum,
maximum);
        }
        if(option==3){
            venus(speed,&minimum, &maximum);
            printf("It will take you %0.2lf to %0.2lf hours to reach
the Venus from Earth at the speed of %0.2lf  km/h.\n\n", speed,
minimum, maximum);
        }
    } while (option!=4);
    printf("Program exited.");
    return (0);
}
```

Output:

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon

```
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 100
It will take you 100.00 to 3631.04 hours to reach the Moon from Earth at the speed of 4056.96
km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 500
It will take you 500.00 to 726.21 hours to reach the Moon from Earth at the speed of 811.39
km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 41000
It will take you 41000.00 to 8.86 hours to reach the Moon from Earth at the speed of 9.90 km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 4
Program exited.
```

Screenshot:

```
Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 100
It will take you 100.00 to 3631.04 hours to reach the Moon from Earth at the speed of 4056.96 km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 500
It will take you 500.00 to 726.21 hours to reach the Moon from Earth at the speed of 811.39 km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 1
Enter your speed in kilometers per hour: 41000
It will take you 41000.00 to 8.86 hours to reach the Moon from Earth at the speed of 9.90 km/h.

Pick one of the following options by entering the number corresponding to the option:
1. Travel to the Moon
2. Travel to Mars
3. Travel to Venus
4. Exit Program
Option: 4
Program exited.

...Program finished with exit code 0
Press ENTER to exit console.
```

Problem #3:
Algorithm:

1. Ask the user to enter each parameter
2. Make sure each parameter inputted is valid
3. Calculate and output the surface area of the trapezoid

Code:

```
#include <stdio.h>

//void function that will calculate the surface area of the trapezoid
void calculate_surface_area(double a, double b, double h, double
*area) {
    *area=0.5*(a+b)*h; //area of trapezoid formula
}

int main(void) {
    //initializing variables
    double a, b, h, area;
```

```
    //ask the user to input the values and validate the input
    do {
        printf("Enter the length of the short base (a): ");
        scanf("%lf", &a);
        if (a<=0){
            printf("Length must be greater than 0.\n");
        }
    } while (a<=0);

    do {
        printf("Enter the length of the long base (b): ");
        scanf("%lf", &b);
        if (b<=0){
            printf("Length must be greater than 0.\n");
        }
    } while (b<=0);

    do {
        printf("Enter the height of the trapezoid (h): ");
        scanf("%lf", &h);
        if (h<=0){
            printf("Height must be greater than 0.\n");
        }
    } while (h<=0);

    //call void function
    calculate_surface_area(a, b, h, &area);

    //output the surface area
    printf("The surface area of the trapezoid is: %0.3lf\n", area);

    return (0);
}
```

Output:
Enter the length of the short base (a): 10
Enter the length of the long base (b): 2
Enter the height of the trapezoid (h): 4

The surface area of the trapezoid is: 24.000

Screenshot:

```
Enter the length of the short base (a): 10
Enter the length of the long base (b): 2
Enter the height of the trapezoid (h): 4
The surface area of the trapezoid is: 24.000


...Program finished with exit code 0
Press ENTER to exit console.
```